

Absolute Bioavailability and Pharmacokinetic Comparability of Sirukumab Following Subcutaneous Administration by a Prefilled Syringe or an Autoinjector

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Abstract

This phase I, randomized, open-label study assessed the absolute bioavailability and pharmacokinetic comparability of sirukumab, a human anti-interleukin-6 monoclonal antibody, following subcutaneous (SC) administration via Prefilled Syringe-UltraSafe Passive[®] Delivery System (PFS-U) or Prefilled Syringe-SmartJect[®] Autoinjector (PFS-AI; Janssen Research & Development, LLC, Spring House, Pennsylvania). A total of 144 healthy male subjects were randomized to 5 single-dose treatment groups: sirukumab 50 mg and 100 mg (each by PFS-U and PFS-AI) and sirukumab 100 mg intravenous (IV) infusion. Pharmacokinetic parameters were calculated using noncompartmental analysis. Following SC administration, maximum serum concentrations (C_{max}) and area under the concentration-vs-time curve (AUC) increased in an approximately dose-proportional manner. Median time to reach C_{max} was 5 days, and mean half-life ranged from 16 to 19 days. Mean absolute bioavailability of sirukumab by PFS-AI and PFS-U, respectively, was estimated at 92.4% and 81.4% with 100 mg and 88.4% and 94.7% with 50 mg. Ratios of geometric means (90% confidence intervals) of C_{max} and AUC_{0-77d} for PFS-AI:PFS-U were 1.13 (1.03, 1.25) and 1.14 (1.05, 1.24), respectively, indicating comparable systemic exposures of sirukumab following a single 100-mg SC dose by PFS-U or PFS-AI. The incidence of antibodies to sirukumab was low (1.4%). No new safety concerns associated with sirukumab were identified at either dose.

Keywords

autoinjector, comparability, pharmacokinetics, sirukumab, subcutaneous administration

Sirukumab (formerly known as CNTO 136), an interleukin-6 (IL-6) inhibitor, is an IgG1 κ human monoclonal antibody that binds specifically to the cytokine IL-6 and is currently under development for the treatment of rheumatoid arthritis (RA) and other diseases.^{1–3} The interaction of sirukumab with IL-6 prevents IL-6 from binding to soluble and membrane-bound IL-6 receptors⁴ and is believed to inhibit IL-6-mediated trans- and classic signaling, respectively. The phase 3 SIRROUND studies that were designed to assess the efficacy and safety of 2 different subcutaneous (SC) dosing regimens of sirukumab (50 mg every 4 weeks and 100 mg every 2 weeks) in patients with RA have completed primary efficacy analyses, and the final publications of these study results are eagerly awaited.

Sirukumab has been administered SC using a prefilled syringe (PFS) fitted with the UltraSafe Passive[®] Delivery System (PFS-U) in phase 3 studies, and a

PFS fitted with the SmartJect[®] Autoinjector (PFS-AI; Janssen Research & Development, LLC, Spring House, Pennsylvania) was available and used in a long-term extension study. Both presentations used the identical prefilled syringe with a fixed 26 G ½-inch needle and had the same liquid formulation and volume of sirukumab. Both of these devices permit at-home administration by patients or their caregivers.⁵ The autoinjector presentation has an easy-to-grip oval body that administers medication with a push of the button

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on the side; after injection, the needle and syringe retract permanently into the device. By contrast, the PFS-U may be more difficult for patients to grip, the plunger must be pushed from the top, and the needle remains outside the device. In several studies that examined patient preferences, the majority of patients reported that they preferred autoinjector administration compared with their prior device or nurse-administered SC injections with a syringe, and that autoinjector administration was associated with lower levels of injection-related pain and discomfort.⁶⁻⁹ Thus, these features may make the autoinjector more desirable as a delivery system for sirukumab than the prefilled syringe.

This phase 1 study was designed to bridge the pharmacokinetics of sirukumab following SC delivery by autoinjector or prefilled syringe in healthy male subjects. The objectives of the study were to assess the absolute bioavailability of sirukumab following SC administration of 50 mg or 100 mg and to evaluate the pharmacokinetic comparability of a single dose of sirukumab 100 mg SC delivered by autoinjector or prefilled syringe.

Methods

Subjects

Healthy men aged 18 to 55 years with a body weight of 50 to 100 kg were eligible for enrollment into the study. Inclusion and exclusion criteria covered the usual standards for clinical trials in healthy volunteers, excluding comorbidities, comedication, and other clinically significant abnormalities and/or potential interfering situations. Subjects were excluded if they were taking any prescription, over-the-counter, herbal, or natural medication, with the exception of acetaminophen and pre-existing multivitamins.

During the study period, subjects were instructed not to use any medication, as per exclusion criteria, and they could not receive any live or attenuated vaccines. Alcohol consumption was not permitted for ≥ 48 hours prior to study drug administration and during the inpatient period of the study. Caffeine-rich beverages or foods were not permitted for 48 hours prior to study drug administration and 4 hours prior to an electrocardiogram (ECG).

Study Design

This was a randomized, open-label, parallel-group, multicenter, single-dose, phase 1 study (Figure 1). The study was conducted at 2 sites in the United States (Celerion, Inc, Tempe, Arizona and Celerion, Inc, Lincoln, Nebraska). Eligible subjects were randomized 3:3:3:7:7 to 1 of the following 5 treatment groups, respectively: sirukumab 100 mg as an intravenous (IV)

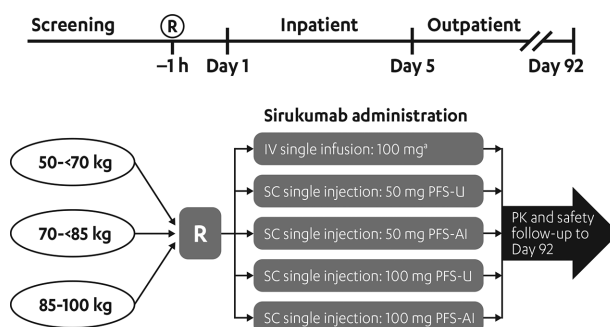


Figure 1. Study design. R, randomization; IV, intravenous; SC, subcutaneous; PFS-U, Prefilled Syringe-UltraSafe Passive[®] Delivery System; PFS-AI, Prefilled Syringe-SmartJect[®] Autoinjector; PK, pharmacokinetics. ^aInfusion over 10 minutes.

infusion, sirukumab 50 mg administered by PFS-AI, sirukumab 50 mg administered by PFS-U, sirukumab 100 mg administered by PFS-AI, and sirukumab 100 mg administered by PFS-U. All medications were administered by qualified study-site personnel. The SC doses were supplied in a 1-mL prefilled syringe and administered in an area on the front of the mid-thigh. The IV infusion was given at a constant rate over approximately 10 minutes in the opposite arm from that used for obtaining pharmacokinetic samples. Randomization was stratified by body weight into 3 categories: 50 to < 70 kg, 70 to < 85 kg, and 85 to 100 kg. The study included a screening period of ≤ 4 weeks, an inpatient period of approximately 1 week, and a follow-up period of ≤ 13 weeks, for a total duration of approximately 18 weeks (Figure 1).

The study was conducted in accordance with the Declaration of Helsinki and consistent with Good Clinical Practices, and all subjects provided written informed consent. The study protocol was reviewed by an institutional review board (Chesapeake Research Review, Inc, Columbia, Maryland).

Pharmacokinetic Assessment

Blood samples were collected from all subjects to measure serum sirukumab concentrations predose and at 0.5 (IV only), 4 (IV only), and 12 hours after administration and at days 2, 3, 4, 5, 6, 7, 15, 22, 36, 50, 64, 78, and 92 after administration. The samples were centrifuged at room temperature and transferred to cryovials for freezing until analysis. As previously described,¹⁰ serum concentrations of sirukumab were determined using a validated, specific, and sensitive immunoassay method (Janssen Research & Development, LLC), with a lower limit of quantification (LLOQ) of 0.0977 $\mu\text{g/mL}$.

Pharmacokinetic parameters were calculated via noncompartmental analysis using PhoenixTM

WinNonlin® (version 6.2.1; Pharsight, A Certara™ Company, St. Louis, Missouri). All calculations were based on actual sampling times. Serum sirukumab concentrations that were below LLOQ were assigned a value of 0 if they preceded the first quantifiable sample in the profile, and any other values below LLOQ were set to missing. Area under the concentration-vs-time curve (AUC) values were calculated using a combination of linear and logarithmic trapezoidal methods, with the linear method applied before the time of maximum serum concentration (T_{max}) and the logarithmic method applied after T_{max} . Log-linear least-squares regression analysis was used to calculate the terminal rate constant (λ_z). The absolute bioavailability (F) after SC administration of 50 mg or 100 mg was calculated as AUC from time 0 to infinity with extrapolation of the terminal phase (AUC_{∞}) following SC administration divided by the mean AUC_{∞} following IV administration $\times 100\%$. For the estimation of F for 50-mg SC arms, 50-mg IV AUC_{∞} was assumed to be 50% of the mean of 100-mg IV AUC_{∞} based on the dose-proportional pharmacokinetics of sirukumab. Values for terminal half-life ($t_{1/2}$), AUC from time 0 to 77 days (AUC_{0-77d}), AUC_{∞} , clearance (CL), and volume of distribution (V_z) were reported only if the adjusted coefficient of determination (R^2) values were ≥ 0.90 and the extrapolated AUC after the last quantifiable serum concentration must not have exceeded 20% of the AUC_{∞} value.

Immunogenicity Assessment

Serum samples were collected at days 0 and 92 for analysis of antibodies to sirukumab. Detection and characterization of antibodies to sirukumab were performed using a sensitive and drug-tolerant electrochemiluminescent immunoassay (ECLIA; Janssen Research & Development, LLC), which included an acid dissociation step to improve detection of antibodies to sirukumab in the presence of excess sirukumab in human serum. The sensitivity of the ECLIA method was 6.2 ng/mL of antibodies to sirukumab in human serum that did not contain sirukumab. As validated, 75 ng/mL of antibodies to sirukumab could be detected in the presence of up to 50,000 ng/mL of sirukumab in human serum. Subjects were classified as positive for antibodies to sirukumab if ≥ 1 sample had detectable antibodies to sirukumab.

Safety

Safety assessments included clinical laboratory tests, ECGs, vital sign measurements, physical examinations, and monitoring of adverse events (AEs), which included injection-site reactions and infusion-related reactions.

Statistical Analyses

No formal hypothesis testing was conducted. Sample size was chosen based on the empirical convention for assessing absolute bioavailability and pharmacokinetic comparability following a single-dose administration in healthy subjects. For all subjects who received study drug, data were summarized using descriptive statistics. Serum concentrations that were below LLOQ were assigned a value of 0 for computing descriptive statistics.

The evaluable pharmacokinetic population included all subjects who received a full dose of study drug and whose pharmacokinetic profiles allowed accurate calculation of ≥ 1 pharmacokinetic parameter. Pharmacokinetic parameters were summarized by treatment group. Because sirukumab has a long $t_{1/2}$, maximum observed serum concentrations (C_{max}) and partial AUC (AUC_{0-77d}) were used as the primary parameters in the evaluation of pharmacokinetic comparability. The pharmacokinetic comparability between PFS-AI and PFS-U was evaluated by comparing geometric means and 90% confidence intervals (CIs) of C_{max} and AUC_{0-77d} for groups receiving sirukumab 100 mg via these 2 devices.

Results

Subjects

A total of 144 subjects were randomized and received study drug. Of these, 139 subjects completed the study through the final follow-up visit at week 13. Five subjects discontinued the study early, 4 of these subjects were lost to follow-up and 1 subject withdrew for personal reasons. Demographic and baseline characteristics were generally well balanced across all treatment groups (Table 1). All subjects were male, with a mean age of 35 years, and the majority were white (86%). Randomization was stratified by body weight and, thus, the body weight categories were well balanced across treatment groups, with a mean (standard deviation [SD]) body weight for all patients of 77.7 (12.8) kg.

Pharmacokinetics

The evaluable pharmacokinetic population included 143 subjects; 1 subject in the 100-mg PFS-U group had an incomplete dose of sirukumab and was excluded from all pharmacokinetic evaluations. After a single IV infusion of sirukumab 100 mg, serum concentrations declined in a biexponential manner and were superimposable in the terminal phase to those observed after a single SC administration of sirukumab 100 mg (Figure 2A). Mean serum sirukumab concentrations peaked approximately 5 days after SC administration of 50 mg or 100 mg and declined in an approximately monoexponential manner. C_{max} and AUC_{∞} increased in an approximately dose-proportional manner after

Table 1. Demographic and Baseline Characteristics

Characteristic	Sirukumab SC				Sirukumab IV 100 mg	Total
	50 mg PFS-AI	50 mg PFS-U	100 mg PFS-AI	100 mg PFS-U		
Subjects treated	19	19	43	44	19	144
Age, mean (SD), y	37.1 (10.3)	34.3 (12.2)	33.8 (9.1)	35.0 (12.1)	37.6 (10.0)	35.1 (10.7)
Weight, mean (SD), kg	78.8 (14.5)	78.8 (12.5)	77.9 (12.4)	76.7 (13.3)	77.8 (12.4)	77.7 (12.8)
Race, n (%) ^a						
White	15 (78.9)	15 (78.9)	37 (86.0)	42 (95.5)	15 (78.9)	124 (86.1)
Black	1 (5.3)	4 (21.1)	5 (11.6)	2 (4.5)	4 (21.1)	16 (11.1)
Asian	2 (10.5)	0	0	0	0	2 (1.4)
American Indian/Alaska Native	1 (5.3)	0	1 (2.3)	0	0	2 (1.4)
Hispanic/Latino, n (%)	11 (57.9)	4 (21.1)	29 (67.4)	26 (59.1)	10 (52.6)	80 (55.6)

SC, subcutaneous; IV, intravenous; PFS-AI, Prefilled Syringe-Smartject® Autoinjector; PFS-U, Prefilled Syringe-UltraSafe Passive® Delivery System; SD, standard deviation.

^aMay not total 100.0% due to rounding.

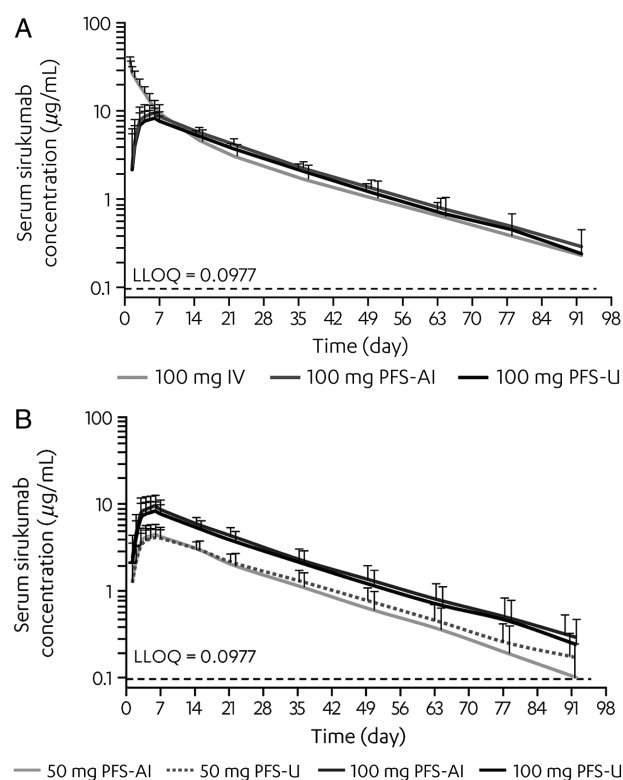


Figure 2. Mean (SD) serum sirukumab concentration over time after (A) a single IV or SC administration of sirukumab 100 mg and (B) a single SC administration of sirukumab 50 mg or 100 mg. SD, standard deviation; IV, intravenous; SC, subcutaneous; LLOQ, lower limit of quantification; PFS-AI, Prefilled Syringe-Smartject® Autoinjector; PFS-U, Prefilled Syringe-UltraSafe Passive® Delivery System.

single SC 50-mg or 100-mg doses, and concentration-vs-time profiles were parallel between the 2 dose groups (Figure 2B). Linear pharmacokinetics were observed for both the 50-mg and 100-mg doses. The coefficient

of variation for C_{max} and AUC_{∞} values ranged from 27% to 35% and 29% to 36%, respectively. The mean $t_{1/2}$ ranged from approximately 16 to 19 days across single SC and IV doses (Table 2).

Absolute Bioavailability

Following SC administration of a 100-mg dose, the mean absolute bioavailability of sirukumab was estimated at 92.4% when administered by PFS-AI and 81.4% when administered by PFS-U. For SC administration of a 50-mg dose, mean absolute bioavailability by PFS-AI and PFS-U administration was 88.4% and 94.7%, respectively.

Pharmacokinetic Comparability of PFS-U and PFS-AI

The ratios of the geometric means (90% CIs) for C_{max} and AUC_{0-77d} from the 100-mg PFS-AI group to those from the 100-mg PFS-U group were 1.13 (1.03, 1.25) and 1.14 (1.05, 1.24), respectively (Table 3). These 90% CIs were within the range of 0.80 to 1.25. For the 50-mg dose, the mean C_{max} for the PFS-AI group (4.64 µg/mL) was similar to that for the PFS-U group (4.40 µg/mL), as was the mean AUC_{0-77d} (110.26 and 115.38 µg·day/mL, respectively). T_{max} was similar for both doses of sirukumab following SC delivery by PFS-AI or PFS-U (median T_{max} was 5 days for both doses and presentations).

Immunogenicity

Among the 19 subjects who received IV sirukumab, 18 had appropriate serum samples for detection of antibodies to sirukumab. None of those 18 subjects tested positive for antibodies to sirukumab after a single 100-mg IV dose.

Of the 125 subjects who received sirukumab SC, 122 had appropriate serum samples for antibody testing.

Table 2. Summary of Sirukumab Pharmacokinetic Parameter Estimates^a

Parameter	Sirukumab SC				Sirukumab IV 100 mg
	50 mg PFS-AI	50 mg PFS-U	100 mg PFS-AI	100 mg PFS-U	
Subjects evaluable	19	19	43	43	19
C_{max} , $\mu\text{g/mL}$	4.64 (1.34)	4.40 (1.20)	9.93 (3.49)	8.68 (2.79)	34.61 (6.35)
T_{max} , days ^b	5 (3-14)	5 (2-14)	5 (1-14)	5 (2-6)	—
AUC_{∞} ($\mu\text{g}\cdot\text{day/mL}$)	116.13 (42.28)	124.35 (35.78)	242.71 (71.07)	213.77 (66.12)	262.70 (77.23)
AUC_{0-77d} ($\mu\text{g}\cdot\text{day/mL}$)	110.26 (36.65)	115.38 (30.12)	228.50 (61.08)	201.58 (56.70)	—
$t_{1/2}$, days	15.87 (4.58)	18.72 (4.61)	17.81 (3.43)	17.13 (4.08)	17.39 (4.62)
CL, mL/day/kg	—	—	—	—	5.43 (1.80)
V_z , mL/kg	—	—	—	—	127.43 (19.54)
F, %	88.42 (32.19)	94.67 (27.24)	92.39 (27.05)	81.38 (25.17)	—

SC, subcutaneous; IV, intravenous; PFS-AI, Prefilled Syringe-Smartject[®] Autoinjector; PFS-U, Prefilled Syringe-UltraSafe Passive[®] Delivery System; C_{max} , maximum observed serum concentration; T_{max} , time to reach maximum observed serum concentration; AUC_{∞} , area under the serum concentration-vs-time curve from time 0 to infinity with extrapolation of the terminal phase; AUC_{0-77d} , area under the serum concentration-vs-time curve from time 0 to 77 days; $t_{1/2}$, half-life; F, absolute bioavailability; SD, standard deviation; CL, clearance; V_z , volume of distribution.

^aValues are mean (SD) unless otherwise noted.

^bMedian (range).

Table 3. Statistical Comparisons of Sirukumab C_{max} and AUC_{0-77d} Between PFS-U and PFS-AI After a Single SC Administration of Sirukumab 100 mg

	Sirukumab SC 100 mg	
	PFS-U	PFS-AI
Subjects evaluable	43	43
C_{max} ($\mu\text{g/mL}$)		
Geometric mean ^a	8.28	9.38
Ratio of geometric means (PFS-AI/PFS-U) ^a		1.13
90% CI ^a		(1.03, 1.25)
AUC_{0-77d} ($\mu\text{g}\cdot\text{day/mL}$) ^b		
Geometric mean ^a	193.85	221.36
Ratio of geometric means (PFS-AI/PFS-U) ^a		1.14
90% CI ^a		(1.05, 1.24)

C_{max} , maximum observed serum concentration; AUC_{0-77d} , area under the serum concentration-vs-time curve from time 0 to 77 days; PFS-U, Prefilled Syringe-UltraSafe Passive[®] Delivery System; PFS-AI, Prefilled Syringe-Smartject[®] Autoinjector; SC, subcutaneous; CI, confidence interval.

^aCalculated by 2-way analysis of variance methods adjusted by body weight category.

^b42 patients were evaluable in each treatment group for AUC_{0-77d} .

Two subjects (1.6%, 2/122) were positive for antibodies to sirukumab, with a titer of 1:160 for 1 subject in the 50-mg PFS-AI group and 1:40 for 1 subject in the 100-mg PFS-U group. Sirukumab pharmacokinetics did not appear to be different for the 2 subjects who tested positive for antibodies to sirukumab compared with the pharmacokinetics of the remaining subjects who tested negative.

Safety

Treatment-emergent AEs (TEAEs) occurred in more subjects in the IV group compared with the SC treatment groups (Table 4). The most frequently reported TEAEs were injection-site erythema (9.7%) and headache (6.9%). The majority of TEAEs were mild in

intensity. Four TEAEs were severe in intensity, and all 4 were considered serious AEs: 1 subject in the 50-mg PFS-AI group had cellulitis, and 1 subject in the 100-mg PFS-AI group reported anxiety, depression, and suicidal ideation. In both cases the subjects had a history of the reported events. No additional AEs of clinical interest (eg, malignancies, active tuberculosis) were reported during this study.

Injection-site reactions were reported in 21 (16.8%) subjects after SC administration of sirukumab, including 6 subjects in the 100-mg PFS-AI group, 9 subjects in the 100-mg PFS-U group, and 3 subjects each in the 50-mg PFS-AI and 50-mg PFS-U groups; no infusion-related reactions were reported after IV infusion of sirukumab 100 mg. All injection-site reactions

Table 4. TEAEs by MedDRA Reported by $\geq 7\%$ of Subjects in Any Treatment Group

TEAE, n (%)	Sirukumab SC				Sirukumab IV 100 mg (n = 19)	Total (n = 144)
	50 mg PFS-AI (n = 19)	50 mg PFS-U (n = 19)	100 mg PFS-AI (n = 43)	100 mg PFS-U (n = 44)		
Subjects with ≥ 1 AE	6 (31.6)	9 (47.4)	22 (51.2)	20 (45.5)	11 (57.9)	68 (47.2)
General or administration site						
Injection-site erythema	2 (10.5)	3 (15.8)	4 (9.3)	5 (11.4)	0	14 (9.7)
Injection-site induration	0	2 (10.5)	0	4 (9.1)	0	6 (4.2)
Nervous system						
Headache	1 (5.3)	2 (10.5)	5 (11.6)	2 (4.5)	0	10 (6.9)
Musculoskeletal/connective tissue						
Pain in extremity	1 (5.3)	0	3 (7.0)	1 (2.3)	0	5 (3.5)

TEAE, treatment-emergent adverse event; MedDRA, Medical Dictionary for Regulatory Activities; SC, subcutaneous; IV, intravenous; PFS-AI, Prefilled Syringe-Smartject® Autoinjector; PFS-U, Prefilled Syringe-UltraSafe Passive® Delivery System.

were mild in intensity, with 1 exception of injection-site pain (moderate intensity) in a subject who received sirukumab 100 mg via PFS-AI. No complaints about difficulty of injection using either SC administration device were reported.

No clinically significant effects of sirukumab were observed on clinical laboratory evaluations (including neutrophil, lipid, and platelet counts and liver function tests), vital sign measurements, physical examination findings, or ECGs. No new safety findings, compared with AEs previously reported for sirukumab, were reported at either the 50-mg or 100-mg doses delivered by either the PFS-AI or the PFS-U.

Discussion

Results of this single-dose, phase 1 study demonstrated that the pharmacokinetics of sirukumab following SC administration were similar between the PFS-U and PFS-AI presentations. The absolute bioavailability of sirukumab following SC administration of 100 mg by PFS-AI and PFS-U was estimated to be 92.4% and 81.4%, respectively. Systemic drug exposure, as measured by C_{max} and AUC, was comparable following sirukumab 100 mg SC delivery by PFS-AI or PFS-U. Delivery by PFS-AI resulted in slightly higher drug exposure (13% to 14%) compared with the PFS-U, as indicated by the geometric mean ratios for C_{max} (1.13) and AUC_{0-77d} (1.14); nevertheless, bioequivalence was demonstrated for delivery by PFS-AI and PFS-U because the 90% CIs for these ratios fell within the range of 0.80 to 1.25. Likewise, for the 50-mg dose, the means for C_{max} and AUC_{0-77d} were similar for the PFS-AI and PFS-U groups. These results mirror those observed with another biologic agent, golimumab, when administered SC by standard needle and syringe or a similar autoinjector device to that used in the current study.¹¹

The use of autoinjectors among patients with RA, including those with severe hand disability, is associated with a high level of acceptance and increased treatment compliance.¹² Thus, data from the current pharmacokinetic study suggest the potential for switching from the existing PFS-U presentation that was used in the phase 3 efficacy studies to the PFS-AI device for administration of sirukumab.

The pharmacokinetic parameters identified in this study were consistent with prior reports of the pharmacokinetics of sirukumab by IV and SC administration in healthy subjects and patients with RA.^{1,3} Following SC administration of a 50-mg dose in a previous phase 1 pharmacokinetic study, sirukumab was slowly absorbed, reaching peak serum concentrations approximately 5 days after dosing.¹⁰ In the current study, administration of sirukumab 50- and 100-mg doses resulted in peak serum concentrations at a median of 5 days. The slower absorption phase of SC administration likely masks the distribution phase, as has been observed with other monoclonal antibodies.¹¹ The mean half-life for sirukumab has been shown to be consistent for both the SC and IV routes of administration at approximately 16 to 19 days, based on the results of the current and prior studies, including a study in patients with active RA.^{1,3,10}

The overall proportion of subjects who developed antibodies to sirukumab was low (1.4%) in this study. There was a slightly higher number of TEAEs reported in the PFS-U group compared with the PFS-AI group. Injection-site reactions were the most frequently reported TEAEs in this study and were reported by 9 subjects in the PFS-AI groups and 12 subjects in the PFS-U groups; none of these injection-site reactions was serious. The most common injection-site reactions were erythema and induration. No complaints about the difficulty of SC injection with either device were

reported. No new safety concerns associated with sirukumab were identified in this study.

In conclusion, sirukumab showed linear pharmacokinetics at dose levels ranging from 50 mg to 100 mg and high absolute bioavailability following a single SC administration in healthy subjects. The pharmacokinetics of sirukumab were comparable following a single SC administration by PFS-U or PFS-AI. No new safety concerns associated with sirukumab were identified in this healthy male population.

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Declaration of Conflicting Interests and Financial Disclosures

Y. Zhuang, S. J. Marciniak, H. Liu, H. Zhou, H. M. Davis, D. Raible, and Z. Xu are employees of Janssen Research & Development, LLC and are shareholders of Johnson & Johnson. D. E. de Vries is an employee of Janssen Biologics BV and is a shareholder of Johnson & Johnson. F. Leon was an employee of Janssen Research & Development, LLC at the time that the research was conducted.

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